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Piyush Arora

Professional Summary

Graduate researcher with experience spanning academia and industry, focused on building deployable machine learning systems. My work lies at the intersection of multimodal models, speech and vision-language learning, and post-training, with publications at leading AI conferences including INTERSPEECH, EMNLP, and WACV. Rather than waiting for the next breakthrough in AI, I aspire to help create it.

Education

- **Imperial College London** — United Kingdom
  MSc Computing (Artificial Intelligence & Machine Learning); Grade: Distinction (Coursework) — *Sep 2025 - Present*
- **Indian Institute of Technology Jodhpur** — India
  Bachelor of Technology in Computer Science and Engineering; CGPA: 8.88/10 — *Dec 2020 - May 2024*
  - *All India Rank **3316 (IIT-JEE 2020)** among 1.3 million applicants.*
  - *Selected for the prestigious **Mitacs Globalink Research Internship**.*
  - *B.Tech. Thesis: Hybrid Sample Synthesis-Based Debiasing in Limited Data Settings; published at **WACV 2024**.*
  - *Coursework: Deep Learning, Advanced Machine Learning, Natural Language Understanding, Data Structures and Algorithms, Operating Systems, Computer Networks, Probability, Statistics, Stochastic Processes, Linear Algebra.*

Publications

- **P. Arora***, N. Singh*, U. Cappellazzo, S. Petridis, and M. Pantic, *VIB-AVSR: Variational Information Bottleneck for Noise-Robust LLM-Based Audio-Visual Speech Recognition*: **INTERSPEECH 2026 (Oral)**
- **P. Arora***, N. Singh*, V. Diwan*, and P. Mazumder, *BLADE: Bias-Linked Adaptive DEbiasing*: Under Review
- A. Subramanyam*, N. Singh*, **P. Arora***, and A. Mishra, *When Big Models Train Small Ones: Label-Free Model Parity Alignment for Efficient Visual Question Answering with Small VLMs*: **EMNLP 2025**
- **P. Arora*** and P. Mazumder*, *Hybrid Sample Synthesis-Based Debiasing Of Classifier*: **WACV 2024**

* denotes equal contribution

Work Experience

- **Neuphonic** — London, UK
  Machine Learning Intern — *May 2026 - Sep 2026*
  - Worked on post-training and distillation of large audio language models for noise-robust Automatic Speech Recognition (ASR) effective in whispering conditions and real-time offline deployment in noisy environments.
  - Applied **GSPO**-based post-training to ASR models for hotword biasing and improved noise robustness.
  - Extended LLMs to natively process and reason over the audio modality, and optimized low-latency inference pipelines for deployment on edge and GPU-based systems.
- **MetaFusion** — Noida, India
  Research Engineer — *May 2024 - Aug 2025*
  - Built end-to-end fine-tuning and evaluation pipelines for VLM/LLM-based surveillance captioning and attribute prediction in PyTorch; designed attribute-specific cross-attention layers, profiled/optimised bottlenecks, and fine-tuned Microsoft Florence-2, improving F1 score from **84% to 91%**.
  - Shipped an intelligent traffic management system integrating multi-violation detection, vehicle recognition, license plate OCR, and CLIP-based attribute prediction, deployed as containerised microservices on GPU-backed infrastructure.
  - Owned production rollout across Gandhinagar, Vizag, and Maharashtra for real-time monitoring; built automated benchmarking for quality and achieved **50%+** end-to-end throughput improvement.

Research Experience

- **Improving Chain-of-Thought Faithfulness in LLMs** — Imperial College London
  Supervisor: Dr. Oana-Maria Camburu, Dr. María Pérez-Ortiz, Noah Y. Siegel | Code — *Oct 2025 – Sep 2026*
  - Developed **I-FAITH**, a reinforcement learning algorithm that uses counterfactual hint interventions to train LLMs to identify and verbalize factors causally influencing their predictions, addressing a critical AI safety challenge in Chain-of-Thought (CoT) monitoring.
  - Designed a **frequency-based simulation reward** with an adaptive answer-consistency constraint, mitigating reward hacking and balancing faithful reasoning against task accuracy using **LUSPO**-based policy optimization.
  - Evaluated on **Qwen3-4B, Qwen3-8B, and Qwen3-14B** across SNLI, ETHICS, and out-of-distribution loan approval, diabetes diagnosis, and admissions tasks; improved average faithfulness GMean from **29.4% to 58.7%** on OOD tasks and demonstrated generalisation to six unseen hint types.
- **VIB-AVSR for Noise-Robust LLM-Based AVSR** — Imperial College London
  Supervisor: Dr. Umberto Cappellazzo, Dr. Stavros Petridis, Dr. Maja Pantic — *Nov 2025 – Mar 2026*

- Developed **VIB-AVSR**, an LLM-based audio-visual speech recognition model connecting pre-trained audio-visual encoders to an LLM backbone, achieving strong performance in clean acoustic conditions.
- Integrated Variational Information Bottleneck layers at targeted positions within the LLM backbone to regularize representations, addressing the lack of explicit noise-robustness mechanisms in existing LLM-based AVSR models.
- Reduced performance degradation across multiple SNR levels and noise types without architectural modifications or additional training data, while maintaining efficient GPU inference across varying noise conditions.

- **Matryoshka-Based Neural Codec for Adaptive Speech Tokenization** — Imperial College London
  Supervisor: Dr. Mirco Ravanelli, Dr. Umberto Cappellazzo, Dr. Stavros Petridis — *Apr 2026 – Present*
  - Developed *Matry*-FocalCodec, extending FocalCodec with Matryoshka Representation Learning to support multiple codec configurations within a single neural audio codec model, rather than requiring separately trained models for each setting.
  - Evaluated the approach along two axes – code dimension and token rate – comparing each configuration against separately trained FocalCodec baselines, and retained competitive performance across both axes within a single model.
  - Demonstrated that Matryoshka Representation Learning provides configuration flexibility in neural audio codecs without sacrificing performance or requiring a separately trained model per configuration.

- **StyleLip: Cross-Identity Speaking Style Transfer** — Imperial College London
  Supervisors: Dr. Antoni Bigata, Dr. Stavros Petridis, Dr. Maja Pantic — *Oct 2025 – Mar 2026*
  - Developed StyleLip, a few-shot framework for speaking style transfer in audio-driven lip-synchronization, capturing and reproducing person-specific idiosyncratic lip movements across different identities using LoRA layers integrated into a base lip-sync model.
  - Regularized training via synthetically generated multi-identity samples with a shared speaking style to prevent identity leakage while enforcing consistent style transfer, achieving SOTA performance as a plug-and-play solution.
  - Introduced SSCS, a novel quantitative metric to measure speaking style transfer fidelity, addressing the lack of dedicated metrics in the field.

- **When Big Models Train Small Ones** — IIT Jodhpur
  Supervisor: Dr. Anand Mishra (VL2G Lab) | Paper — *Dec 2024 – Mar 2025*
  - Designed **Model Parity Alignment (MPA)** for Visual Question Answering to train small VLMs (SmolVLM-500M, TinyLLaVA-2B, InternVL2-2B/4B) from large VLMs (Qwen2VL-7B, InternVL2-8B, GPT-4o) using unlabeled data, implementing efficient PyTorch training pipelines across multiple model scales.
  - Developed disparity-aware training to mitigate hallucinations and close reasoning gaps between VLM and its larger counterpart via automated QA pair generation and filtering of disparity samples to guide fine-tuning across OCR, commonsense, and factual tasks.
  - Achieved consistent accuracy gains up to **6%** on VQA benchmarks like TextVQA, ST-VQA, OKVQA, MedicalVQA, and ChartQA.

- **Hybrid Sample Synthesis-Based Debiasing in Limited Data Settings** — IIT Jodhpur
  Supervisor: Dr. Pratik Mazumder | Paper — *Dec 2022 – Mar 2023*
  - Designed a hybrid sample synthesis method for low-data regimes, enabling ResNet classifiers to generalize under unknown biases and promoting AI fairness.
  - Built a dual-model framework to create bias-conflicting samples without relying on explicit bias annotations.
  - Achieved up to **+10%** accuracy improvement over prior methods on benchmarks such as corrupted CIFAR-10, Colored MNIST, and BFFHQ.

Skills Summary

- **Programming**: Python, C++, SQL, Bash
- **ML & AI Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV
- **LLM Post-Training & Evaluation**: Supervised Fine-Tuning (SFT), RL-based training (GRPO), Reward-Driven Optimization, Ablations, Benchmarking
- **NLP & Multimodal**: Transformers, Instruction Tuning, Reasoning Faithfulness; Audio-Visual Modeling
- **Quantitative Skills**: Probability Theory, Statistics, Optimization, Stochastic Processes, Monte Carlo Simulation
- **Tools & Platforms**: Docker, Kubernetes, Git, Linux; experience deploying ML services on GPU-based infrastructure
- **Other**: Technical Writing, Public Speaking, Leadership, Project Management